

PX-57: AB72 + Flux + Lubricant

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Date of issue: 10/23/2018

Revision date: 10/23/2018

Version: 1.0

SECTION 1: Identification

1.1. Identification

Product form : Mixture
Product name : PX-57: AB72 + Flux + Lubricant

1.2. Recommended use and restrictions on use

Use of the substance/mixture : Powder Metallurgy Use

1.3. Supplier

Manufacturer

ARC Metals, Hoeganaes Corporation
PO Box 372
224 River Road
Ridgway, Pa 15853 - USA
T (814) 776-2116
F (814) 776-4000

1.4. Emergency telephone number

Emergency number : INFOTRAC Emergency Response Center
200 North Palmetto Street; Leesburg, Florida 34748
Tel.# 800-535-5053 / 352-323-3500
Fax.# 352-326-3902

SECTION 2: Hazard(s) identification

2.1. Classification of the substance or mixture

GHS-US classification

Reproductive Toxicity, 1B
Combustible Dust

2.2. GHS Label elements, including precautionary statements

GHS-US labeling

Hazard pictograms (GHS-US) :



Signal word (GHS-US) : Danger

Hazard statements (GHS-US) : May form combustible dust concentrations in air
May damage fertility or the unborn child

Precautionary statements (GHS-US) : Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
Wear protective gloves/protective clothing/eye protection/face protection.
If exposed or concerned: Get medical advice/attention.
Store locked up.
Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation

2.3. Other hazards which do not result in classification

No additional information available

2.4. Unknown acute toxicity (GHS US)

Not applicable

SECTION 3: Composition/Information on ingredients

3.1. Substances

Not applicable

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3.2. Mixtures

Name	Product identifier	%
Nickel	(CAS-No.) 7440-02-0	30 - 50
Copper	(CAS-No.) 7440-50-8	30 - 40
Manganese	(CAS-No.) 7439-96-5	10 - 20
Boric acid (H3BO3)	(CAS-No.) 10043-35-3	1 - 7
Disodium tetraborate	(CAS-No.) 1330-43-4	< 5
Silicon	(CAS-No.) 7440-21-3	1 - 5
Boron	(CAS-No.) 7440-42-8	1 - 5

*Chemical name, CAS number and/or exact concentration have been withheld as a trade secret

SECTION 4: First-aid measures

4.1. Description of first aid measures

- First-aid measures after inhalation : If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical advice/attention if you feel unwell.
- First-aid measures after skin contact : If skin irritation occurs: Wash skin with plenty of water. Obtain medical attention if irritation persists.
- First-aid measures after eye contact : IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.
- First-aid measures after ingestion : Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person. Get medical advice/attention if you feel unwell.

4.2. Most important symptoms and effects (acute and delayed)

- Symptoms/effects after inhalation : May cause irritation to the respiratory tract.
- Symptoms/effects after skin contact : May cause skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin.
- Symptoms/effects after eye contact : May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling.
- Symptoms/effects after ingestion : May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.
- Chronic symptoms : Toxic to reproductive function and development.

4.3. Immediate medical attention and special treatment, if necessary

Symptoms may be delayed. In case of accident or if you feel unwell, seek medical advice immediately (show the label or SDS where possible).

SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media : Use extinguishing media appropriate for surrounding fire.
- Unsuitable extinguishing media : Do not use water jet.

5.2. Specific hazards arising from the chemical

- Fire hazard : Combustible dust. Products of combustion may include, and are not limited to: oxides of carbon. Metal oxides.
- Explosion hazard : Airborne dust in sufficient concentrations when confined and exposed to a sufficient ignition source can explode.
- Reactivity : No dangerous reactions known under normal conditions of use.

5.3. Special protective equipment and precautions for fire-fighters

- Protection during firefighting : Keep upwind of fire. Wear full fire fighting turn-out gear (full Bunker gear) and respiratory protection (SCBA).

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

- General measures : Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel. Dust deposits should not be allowed to accumulate on surfaces, as these may form an explosive mixture if they are released into the atmosphere in sufficient concentration. Avoid dispersal of dust in the air (ie, clearing dust surfaces with compressed air). Use only non-sparking tools.

6.1.1. For non-emergency personnel

No additional information available

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6.1.2. For emergency responders

No additional information available

6.2. Environmental precautions

Prevent entry to sewers and public waters.

6.3. Methods and material for containment and cleaning up

- For containment : Contain spill, then place in a suitable container. Minimize dust generation. Do not flush to sewer or allow to enter waterways. Use appropriate Personal Protective Equipment (PPE).
- Methods for cleaning up : Vacuum or sweep material and place in a disposal container. Provide ventilation.

6.4. Reference to other sections

For further information refer to section 8: "Exposure controls/personal protection"

SECTION 7: Handling and storage

7.1. Precautions for safe handling

- Precautions for safe handling : Avoid generating and breathing dust. Avoid contact with skin, eyes and clothing. Do not swallow. Handle and open container with care. When using do not eat, drink or smoke. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Good housekeeping is important to prevent accumulation of dust. The use of compressed air for cleaning clothing, equipment, etc, is not recommended. Handling this product may result in electrostatic accumulation. Use proper grounding procedures. Use only in well ventilated areas.
- Hygiene measures : Wash contaminated clothing before reuse. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

- Storage conditions : Keep out of the reach of children. Keep container tightly closed. Keep away from sources of ignition. Store locked up. Store in dust-tight, dry, labelled containers. Avoid any dust buildup by frequent cleaning and suitable construction of the storage area.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Nickel (7440-02-0)		
ACGIH	ACGIH TWA (mg/m ³)	1.5 mg/m ³ (inhalable particulate matter)
OSHA	OSHA PEL (TWA) (mg/m ³)	1 mg/m ³
IDLH	US IDLH (mg/m ³)	10 mg/m ³
NIOSH	NIOSH REL (TWA) (mg/m ³)	0.015 mg/m ³
Copper (7440-50-8)		
ACGIH	ACGIH TWA (mg/m ³)	0.2 mg/m ³ (fume)
OSHA	OSHA PEL (TWA) (mg/m ³)	0.1 mg/m ³ (fume) 1 mg/m ³ (dust and mist)
IDLH	US IDLH (mg/m ³)	100 mg/m ³ (dust, fume and mist)
NIOSH	NIOSH REL (TWA) (mg/m ³)	1 mg/m ³ (dust and mist) 0.1 mg/m ³ (fume)
Manganese (7439-96-5)		
ACGIH	Local name	Manganese
ACGIH	ACGIH TWA (mg/m ³)	0.02 mg/m ³ (respirable particulate matter) 0.1 mg/m ³ (inhalable particulate matter)
ACGIH	Remark (ACGIH)	CNS impair; A4 (Not classifiable as a Human Carcinogen: Agents which cause concern that they could be carcinogenic for humans but which cannot be assessed conclusively because of a lack of data. In vitro or animal studies do not provide indications of carcinogenicity which are sufficient to classify the agent into one of the other categories)
ACGIH	Regulatory reference	ACGIH 2018
OSHA	OSHA PEL (Ceiling) (mg/m ³)	5 mg/m ³ (fume)
IDLH	US IDLH (mg/m ³)	500 mg/m ³
NIOSH	NIOSH REL (TWA) (mg/m ³)	1 mg/m ³ (fume)

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Manganese (7439-96-5)		
NIOSH	NIOSH REL (STEL) (mg/m ³)	3 mg/m ³
Silicon (7440-21-3)		
OSHA	OSHA PEL (TWA) (mg/m ³)	15 mg/m ³ (total dust) 5 mg/m ³ (respirable fraction)
NIOSH	NIOSH REL (TWA) (mg/m ³)	10 mg/m ³ (total dust) 5 mg/m ³ (respirable dust)
Boron (7440-42-8)		
Not applicable		
Boric acid (H3BO3) (10043-35-3)		
ACGIH	ACGIH TWA (mg/m ³)	2 mg/m ³ (inhalable particulate matter)
ACGIH	ACGIH STEL (mg/m ³)	6 mg/m ³ (inhalable particulate matter)
Disodium tetraborate (1330-43-4)		
ACGIH	ACGIH TWA (mg/m ³)	2 mg/m ³ (inhalable particulate matter)
ACGIH	ACGIH STEL (mg/m ³)	6 mg/m ³ (inhalable particulate matter)
NIOSH	NIOSH REL (TWA) (mg/m ³)	1 mg/m ³

8.2. Appropriate engineering controls

- Appropriate engineering controls : Ensure good ventilation of the work station. It is recommended that all dust control equipment such as local exhaust ventilation and material transport systems involved in handling of this product contain explosion relief vents or an explosion suppression system or an oxygen-deficient environment. Ensure that dust-handling systems (such as exhaust ducts, dust collectors, vessels, and processing equipment) are designed in a manner to prevent the escape of dust into the work area (i.e. there is not leakage from the equipment). Use only appropriately classified electrical equipment and powered industrial trucks.
- Environmental exposure controls : Avoid release to the environment.

8.3. Individual protection measures/Personal protective equipment

Hand protection:

Wear suitable gloves

Eye protection:

Safety glasses or goggles are recommended when using product.

Skin and body protection:

Wear suitable protective clothing

Respiratory protection:

In case of insufficient ventilation, wear suitable respiratory equipment. Respirator selection must be based on known or anticipated exposure levels, the hazards of the product and the safe working limits of the selected respirator.

Other information:

Handle in accordance with good industrial hygiene and safety procedures. Do not eat, drink or smoke when using this product.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

- Physical state : Solid
- Appearance : Fine powder
- Color : Light to dark gray
- Odor : None
- Odor threshold : No data available
- pH : No data available
- Melting point : 930 - 990 °C (1760 - 1814 °F)
- Freezing point : No data available
- Boiling point : No data available

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Flash point	: No data available
Relative evaporation rate (butyl acetate=1)	: No data available
Flammability (solid, gas)	: Combustible dust
Vapor pressure	: No data available
Relative vapor density at 20 °C	: No data available
Relative density	: Undetermined
Solubility	: No data available
Partition coefficient n-octanol/water	: No data available
Auto-ignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity, kinematic	: No data available
Viscosity, dynamic	: No data available
Explosion limits	: No data available
Explosive properties	: No data available
Oxidizing properties	: No data available

9.2. Other information

No additional information available

SECTION 10: Stability and reactivity

10.1. Reactivity

No dangerous reactions known under normal conditions of use.

10.2. Chemical stability

Stable under normal conditions. May form combustible dust concentrations in air.

10.3. Possibility of hazardous reactions

No dangerous reactions known under normal conditions of use.

10.4. Conditions to avoid

Heat. Avoid dust formation. Incompatible materials.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

May include, and are not limited to: oxides of carbon. Metal oxides.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral)	: Not classified.
Acute toxicity (dermal)	: Not classified
Acute toxicity (inhalation)	: Not classified

Nickel (7440-02-0)	
LD50 oral rat	> 9000 mg/kg
LC50 inhalation rat	> 10.2 mg/l (Exposure time: 1 h)
Copper (7440-50-8)	
ATE US (oral)	500 mg/kg body weight
Manganese (7439-96-5)	
LD50 oral rat	9 g/kg
ATE US (oral)	9000 mg/kg body weight
Silicon (7440-21-3)	
LD50 oral rat	3160 mg/kg
ATE US (oral)	3160 mg/kg body weight
Boron (7440-42-8)	
LD50 oral rat	650 mg/kg
ATE US (oral)	650 mg/kg body weight

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Boric acid (H3BO3) (10043-35-3)

LD50 oral rat	2660 mg/kg
LD50 dermal rabbit	> 2000 mg/kg
LC50 inhalation rat	> 0.16 mg/l/4h
ATE US (oral)	2660 mg/kg body weight

Disodium tetraborate (1330-43-4)

LD50 oral rat	2660 mg/kg
LD50 dermal rabbit	> 2000 mg/kg
LC50 inhalation rat	> 2 mg/m ³ (Exposure time: 4 h)
ATE US (oral)	2660 mg/kg body weight
ATE US (vapors)	0.05 mg/l/4h

Skin corrosion/irritation	: Not classified
Serious eye damage/irritation	: Not classified
Respiratory or skin sensitization	: Not classified.
Germ cell mutagenicity	: Not classified.
Carcinogenicity	: Not classified.

Nickel (7440-02-0)

IARC group	2B - Possibly carcinogenic to humans
National Toxicology Program (NTP) Status	Reasonably anticipated to be Human Carcinogen
In OSHA Hazard Communication Carcinogen list	Yes

As demonstrated by the following test protocols, nickel metal diffusion alloyed to the iron contained in these products, is fully bound and is non-leachable in either neutral leachate test conditions or biologically buffered environments. Therefore, these products are believed to be incapable of causing occupational health or respiratory impairment normally associated with nickel under normal conditions of use or as a result of extreme upset and are not classified as such in accordance with HazCom 2012 and/or GHS Revision 3 criteria.

- As demonstrated by GHS-XRC(M) protocols, these products contain fully bound metals which are non-leachable in either neutral leachate test conditions or biologically buffered environments.
- Dynamic Light Scattering and Laser Particle Size Distribution techniques, the particulate present within these products contain no toxicologically relevant respirable fractions (<10µm or <5µm) above non-detectable trace levels.
- Indiana Neutral Leach, with ICP analysis performed in accordance with EPA 1311 (TCLP)/EPA 6100C and Water Leach/pH protocols.
- Bioavailability characterizations determined by leaching in porcine mucous extraction fluid. ICP analysis in accordance with ASTM5550/D5965 Test Method B MOPS Mucin Buffer Leach/EPA 6010C Water Leach/pH protocols.

Reproductive toxicity	: May damage fertility or the unborn child.
Specific target organ toxicity – single exposure	: Not classified
Specific target organ toxicity – repeated exposure	: Not classified.
Aspiration hazard	: Not classified
Symptoms/effects after inhalation	: May cause irritation to the respiratory tract.
Symptoms/effects after skin contact	: May cause skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin.
Symptoms/effects after eye contact	: May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling.
Symptoms/effects after ingestion	: May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.
Chronic symptoms	: Toxic to reproductive function and development.
Other information	: Likely routes of exposure: ingestion, inhalation, skin and eye.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - General	: Not toxic to aquatic life
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As demonstrated by the following test protocols, copper and nickel metals diffusion alloyed to the iron contained in these products, is fully bound and is non-leachable in either neutral leachate test conditions or biologically buffered environments, resulting in no biological hazard classifications from this product.

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- As demonstrated by GHS-XRC(M) protocols, these products contain fully bound metals which are non-leachable in either neutral leachate test conditions or biologically buffered environments.
- Environmental leachate testing was conducted using the Indiana Neutral Leach procedure to determine potential for environmental release (ASTM D3987-85)

SECTION 13: Disposal considerations

Product/Packaging disposal recommendations : Dispose in a safe manner in accordance with local/national regulations.

SECTION 14: Transport information

14.1 Inland Waterways Transport

Not regulated for transportation according to U.S. federal and/or international regulations. Always refer to applicable regulations for further information.

14.2 Marine Transport

Not regulated for transportation according to U.S. federal and/or international regulations. Always refer to applicable regulations for further information.

14.3 Air Transport (ICAO-TI and IATA-DGR)

Not regulated for transportation according to U.S. federal and/or international regulations. Always refer to applicable regulations for further information.

SECTION 15: Regulatory information

15.1. US Federal regulations

All components of this product are listed, or excluded from listing, on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory

15.2. International regulations

No additional information available

15.3. US State regulations

 **WARNING:** This product can expose you to Nickel, which is known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

SECTION 16: Other information

Date of issue : 10/23/2018
Revision date : 10/23/2018
Other information : Refer to NFPA 654, Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids, for safe handling.

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